

Problem Statement – Case-Based Scenario

Case: Phishing URL Detection

A company wants to protect its employees and customers from **phishing websites**. Attackers create fake URLs that look similar to legitimate websites and use them to trick users into providing sensitive information.

The company has a dataset containing different **URL characteristics**, such as URL length, path length, HTTPS status, number of dots, hyphens, special characters, and suspicious words.

The task is to develop a **machine learning system** that can analyze these URL characteristics and classify a URL as either **Legitimate (0)** or **Phishing (1)**.

The project uses **Exploratory Data Analysis (EDA)** to understand the dataset and **Logistic Regression with Ridge and Lasso regularization** to build and evaluate the classification model.

Case Example

Case: College Student Receiving a Scholarship Link

A college student receives an email saying:

“Congratulations! You have been selected for a scholarship. Click here to claim it.”

The student clicks the link, which takes them to a website that looks like a legitimate scholarship portal.

Before the student enters their personal information, our **Phishing URL Detection system** checks the URL's features, such as:

- URL length
- Path length
- HTTPS status
- Number of dots and hyphens
- Special characters
- Sensitive words

The machine-learning model then gives the result:

URL → Feature Analysis → Logistic Regression → Phishing / Legitimate

If the URL is suspicious, the system classifies it as **Phishing (1)** and warns the student.

Objective of the Case Study

The main objective of this case study is to **develop a machine-learning system that can identify whether a URL is legitimate or phishing** by analyzing its URL characteristics.

Specific objectives:

1. Analyze URL features such as length, path length, HTTPS, dots, hyphens, and sensitive words.
2. Perform EDA to understand patterns in legitimate and phishing URLs.
3. Apply Logistic Regression for classification.
4. Use Ridge and Lasso regularization to control overfitting.
5. Evaluate the model using accuracy, precision, recall, F1-score, and confusion matrix.